

# Learning an Agent-Centered Geometry of Quantum Possibility

Predictive State, Goal-Conditioned Control, and Directed Reachability from Measurement Experience Alone

RL Experiments Project  
Author information omitted for review

August 2026

## Abstract

We study a deliberately austere quantum-control problem. An artificial agent may choose a measurement and observe its classical outcome, but it is never given the quantum state, Kraus operators, Born probabilities, Hamiltonian, or tomographic estimates. Its experience is only a sequence of interventions and consequences. The scientific objective is therefore broader than quantum control: what operational representation of an unknown world can be learned from first-person experience, and can many possible aims be organized into a useful geometry?

We compare finite-history tabular Q-learning, goal-conditioned recurrent Q-learning, and a predictive multi-goal gated recurrent unit (GRU). The full agent jointly learns reward value, one-step outcome prediction, a stochastic-shortest-path cost, and goal embeddings regularized by policy similarity. Five environment families vary Hilbert-space dimension, hidden initial state, measurement strength, informational completeness, and outcome alphabet, including a qubit symmetric informationally complete (SIC) measurement and mutually unbiased qutrit bases. Thirty-six agents are trained across six scenarios, three backends, and two seeds.

The predictive geometry agent has the best macro-average success (80.5%), versus 79.1% for finite-history Q-learning and 77.7% for the reward-only GRU, but no backend dominates every environment and neural seed variance is large. Euclidean goal-embedding distance correlates moderately with held-out policy distance (mean Spearman  $\rho = 0.54$ ) and more weakly with independent trajectory distance ( $\rho = 0.31$ ). Learned reachability is directionally informative but not metrically calibrated: blank-history cost correlates  $r = 0.47$  with empirical conditional hitting time and has mean absolute error 1.75 interventions. Intervention maps nevertheless reveal a legible directed structure in which outcomes reduce distance to matching goals while often increasing distance to alternatives. Goal geometry is thus a productive diagnostic and planning language, but its metric interpretation requires stronger validation.

We therefore perform a second, inverse-designed study aimed directly at emergent space. A nine-dimensional system supports nine place goals and eight movement instruments. Optimizing diagonal-move success selects  $p_d = 0.715$ , near  $1/\sqrt{2}$ , and gives exact 2D stress 0.0365. Three coordinate-free Q-learning agents achieve 100% all-pairs navigation; their learned hitting costs have 1D/2D/3D stresses 0.372/0.071/0.064 and recover the concealed  $3 \times 3$  arrangement with Procrustes  $R^2 = 0.975$ . Partial-observability and Manhattan controls identify operational localization and isotropic action cost as necessary ingredients. This constitutes a limited existence proof for stochastic motion in an agent’s emergent hodological space and motivates a spatial-base/internal-fiber research program.

**Keywords:** reinforcement learning; quantum measurement; partial observability; predictive state; goal-conditioned control; reachability; representation learning; QBism; hodological space; multidimensional scaling; emergent space

# Contents

|           |                                                            |           |
|-----------|------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                        | <b>3</b>  |
| <b>2</b>  | <b>Hidden quantum dynamics</b>                             | <b>4</b>  |
| 2.1       | Quantum instruments and backaction . . . . .               | 4         |
| 2.2       | Environment catalog . . . . .                              | 5         |
| <b>3</b>  | <b>Reinforcement learning from first principles</b>        | <b>5</b>  |
| 3.1       | MDPs, return, and value . . . . .                          | 5         |
| 3.2       | Q-learning and exploration . . . . .                       | 5         |
| 3.3       | Partial observability and predictive state . . . . .       | 6         |
| 3.4       | GRU memory and deep temporal-difference learning . . . . . | 6         |
| 3.5       | Evaluation principles . . . . .                            | 6         |
| <b>4</b>  | <b>Sequence goals and three geometries</b>                 | <b>6</b>  |
| 4.1       | Goal-conditioned control . . . . .                         | 6         |
| 4.2       | Symmetric strategic similarity . . . . .                   | 7         |
| 4.3       | Trajectory similarity . . . . .                            | 7         |
| 4.4       | Directed reachability and displacement . . . . .           | 7         |
| <b>5</b>  | <b>Architecture and experimental protocol</b>              | <b>8</b>  |
| <b>6</b>  | <b>Results</b>                                             | <b>9</b>  |
| 6.1       | Control performance . . . . .                              | 9         |
| 6.2       | Goal-geometry validation . . . . .                         | 9         |
| 6.3       | Reachability and intervention displacement . . . . .       | 9         |
| <b>7</b>  | <b>Inverse-designing emergent two-dimensional space</b>    | <b>14</b> |
| 7.1       | Hodological criterion and bilevel objective . . . . .      | 14        |
| 7.2       | Nine-dimensional quantum construction . . . . .            | 14        |
| 7.3       | Outer optimization and learned geometry . . . . .          | 15        |
| <b>8</b>  | <b>Discussion and limitations</b>                          | <b>19</b> |
| <b>9</b>  | <b>Outlook</b>                                             | <b>19</b> |
| <b>10</b> | <b>Conclusion</b>                                          | <b>22</b> |
| <b>A</b>  | <b>Reproduction and artifacts</b>                          | <b>22</b> |
| <b>B</b>  | <b>Goal repertoires</b>                                    | <b>22</b> |

# 1 Introduction

Quantum control is usually described from the standpoint of an external modeler, who writes down a state  $\rho$ , a Hamiltonian, and control or measurement operators before optimizing a protocol. Model-free quantum control instead asks what can be learned through interaction, an approach already used for quantum feedback, state preparation, and error correction [18–20].

This work imposes a stricter informational boundary. The agent chooses an intervention  $a_t$  and receives a classical outcome  $o_t$ . It is not given  $\rho_t$ , measurement operators, expectation values, fidelities, or tomography. Its information at time  $t$  is

$$h_t = (a_1, o_1, \dots, a_{t-1}, o_{t-1}). \quad (1)$$

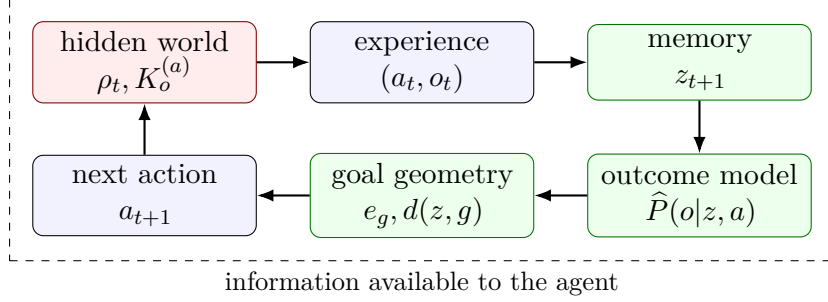
This interface is compatible with QBism’s emphasis on an agent’s actions, experiences, and expectations [1], although the algorithms do not assume or establish any interpretation of quantum theory. Agent-centered here describes the information boundary.

The central difficulty is partial observability. An external physicist may use  $\rho_t$  as a Markov state, but the learner cannot. It needs an internal summary  $z_t = F(h_t)$  retaining distinctions relevant to prediction and control. This motivates three objects that must remain distinct:

1. **predictive state**: which histories imply the same future intervention/outcome statistics;
2. **strategic goal geometry**: which goals call for similar policies or trajectories;
3. **directed reachability**: how difficult a goal is from the agent’s present history.

Our methodological contribution is a validation framework comparing learned Euclidean goal embeddings with held-out policy behavior, trajectory signatures, empirical hitting times, finite-horizon success curves, and intervention-induced changes in all goal distances.

The stronger motivation is hodological: for an agent, goals that require few reliable interventions are near and difficult goals are far. Lewin introduced path-structured “life space” as a formal language for goal-directed behavior [21]. Such a space is generally directed and high-dimensional. Familiar physical space would be a special regime in which a large family of place-like goals admits a common low-stress 2D or 3D embedding and learned action histories become trajectories through it. We treat that as an inverse problem over initial states, Kraus instruments, observation coarse-grainings, and goal repertoires, not as an assumption.



**Figure 1:** The operational learning loop. Quantum variables in red are private to the simulator. The agent constructs state, prediction, and goal-relative reachability from experienced action/outcome pairs.

## 2 Hidden quantum dynamics

### 2.1 Quantum instruments and backaction

A finite-dimensional quantum state is externally represented by a positive semidefinite unit-trace density operator  $\rho$ . A measurement action  $a$  has a Kraus operator  $K_o^{(a)}$  for each classical outcome  $o$ , with physical completeness [2, 3]

$$\sum_o K_o^{(a)\dagger} K_o^{(a)} = I. \quad (2)$$

The hidden simulator samples and updates by

$$p(o|\rho, a) = \text{Tr}(K_o^{(a)} \rho K_o^{(a)\dagger}), \quad (3)$$

$$\rho' = \frac{K_o^{(a)} \rho K_o^{(a)\dagger}}{p(o|\rho, a)}. \quad (4)$$

Measurement therefore both supplies information and changes the conditions governing future experiences. This backaction is why reachability is naturally directed. The learner never evaluates equations (3) and (4); it receives only the sampled integer outcome.

For a projective basis  $\{|b_o\rangle\}$ , one may take  $K_o = |b_o\rangle\langle b_o|$ . The qubit worlds use Pauli  $Z$ ,  $X$ , and  $Y$  bases. Their noncommutativity creates incompatible preparation strategies. The unsharp binary instruments use axis projectors  $P_+$ ,  $P_-$  and strength  $q \in (1/2, 1)$ :

$$K_0 = \sqrt{q} P_+ + \sqrt{1-q} P_-, \quad K_1 = \sqrt{1-q} P_+ + \sqrt{q} P_-. \quad (5)$$

At  $q = 0.5$  the outcome is uninformative; as  $q \rightarrow 1$  the operation approaches a projective measurement. We use  $q = 0.8$ .

A SIC-POVM in dimension  $d$  contains  $d^2$  equiangular rank-one effects and is informationally complete [4]. For the qubit tetrahedral SIC,

$$E_i = \frac{I + \mathbf{r}_i \cdot \boldsymbol{\sigma}}{4}, \quad \sum_{i=0}^3 E_i = I, \quad (6)$$

where  $\mathbf{r}_i$  are tetrahedral Bloch vectors; we use the positive square-root instrument  $K_i = \sqrt{E_i}$ . This tests mixed outcome alphabets: projective actions have two outcomes while the SIC has four. Invalid logits are masked.

Two bases  $\{|e_j\rangle\}$  and  $\{|f_k\rangle\}$  are mutually unbiased when  $|\langle e_j | f_k \rangle|^2 = 1/d$  for all  $j, k$ . Certainty in one becomes a uniform distribution in the other. MUBs formalize complementarity and support tomography [5]. The qutrit world uses three projective MUBs.

**Table 1:** World/state scenarios. The hidden state is reset each episode.

| Environment     | $d$ | Initial state | Measurement actions                    | Goals |
|-----------------|-----|---------------|----------------------------------------|-------|
| qubit-zx-weak   | 2   | $ 1\rangle$   | projective $Z, X$ ; weak $Z$           | 7     |
| qubit-zx-weak   | 2   | $ +\rangle$   | projective $Z, X$ ; weak $Z$           | 7     |
| qubit-pauli     | 2   | $ +i\rangle$  | projective $Z, X, Y$                   | 7     |
| qubit-unsharp   | 2   | $I/2$         | weak $Z, X, Y$                         | 6     |
| qubit-pauli-sic | 2   | $I/2$         | projective $Z, X, Y$ ; tetrahedral SIC | 9     |
| qutrit-mub      | 3   | $ 2\rangle$   | three projective MUBs                  | 8     |

## 2.2 Environment catalog

Operational state is relative to the available interventions. With an informationally incomplete repertoire, physically different density operators may imply identical statistics for everything an agent can do. Adding a SIC action or moving from qubit to qutrit may require a richer predictive representation. This motivates varying worlds rather than only random seeds.

## 3 Reinforcement learning from first principles

### 3.1 MDPs, return, and value

Reinforcement learning concerns sequential decisions with delayed consequences [6]. A finite Markov decision process is  $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ : states, actions, a transition/reward law, and discount  $\gamma \in [0, 1]$ . The Markov property says the present state contains all past information needed to predict the next transition. A policy  $\pi(a|s)$  is a distribution over actions. The discounted return is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \quad (7)$$

The action value is  $Q^\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$ . Optimal value obeys the Bellman equation

$$Q^*(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} Q^*(S_{t+1}, a') | s, a]. \quad (8)$$

The value of acting now is immediate reward plus the best discounted value available afterward.

### 3.2 Q-learning and exploration

Q-learning estimates [equation \(8\)](#) without knowing  $P$  or  $R$  [7]. From transition  $(s_t, a_t, r_t, s_{t+1})$ , it forms target  $y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a')$  (or  $r_t$  at termination) and updates

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[y_t - Q(s_t, a_t)]. \quad (9)$$

The bracket is the temporal-difference error. Q-learning is off-policy: the target assumes a greedy continuation even when behavior explores. With  $\epsilon$ -greedy behavior, a uniformly random action is chosen with probability  $\epsilon$  and a maximizing action otherwise. Classical convergence guarantees require finite tabular state/action spaces, suitable learning rates, and sufficient visitation [7]; they do not automatically extend to nonlinear recurrent networks.

### 3.3 Partial observability and predictive state

Here the simulator’s  $\rho_t$  could be Markov, but is hidden. The problem is a POMDP [8]; the policy is  $\pi(a_t|h_t)$ , not  $\pi(a_t|\rho_t)$ . The tabular baseline truncates experience to

$$s_t^{(L)} = (g, j_t, (a_{t-L}, o_{t-L}), \dots, (a_{t-1}, o_{t-1})), \quad (10)$$

where  $g$  is the goal and  $j_t$  its progress. This is transparent, but  $L$  is arbitrary, its state count grows exponentially, and similar histories share no strength unless identical.

Predictive-state representations instead characterize state by action-conditional predictions [9]. Ideally,

$$h \sim_{\text{pred}} h' \iff \mathbb{P}(o_{1:k}|a_{1:k}, h) = \mathbb{P}(o_{1:k}|a_{1:k}, h') \quad (11)$$

for every future test. Our recurrent vector is only an approximation, trained for control and one-step prediction. The predictive-state view supplies the interpretation and a future validation standard.

### 3.4 GRU memory and deep temporal-difference learning

A recurrent network maintains fixed-size memory

$$z_{t+1} = F_\theta(z_t, a_t, o_t). \quad (12)$$

With  $x_t$  the concatenated action/outcome embedding, a GRU [10] can be written

$$r_t = \sigma(W_r x_t + U_r z_t + b_r), \quad u_t = \sigma(W_u x_t + U_u z_t + b_u), \quad (13)$$

$$\tilde{z}_{t+1} = \tanh(W_h x_t + U_h(r_t \odot z_t) + b_h), \quad z_{t+1} = (1 - u_t) \odot \tilde{z}_{t+1} + u_t \odot z_t. \quad (14)$$

Reset and update gates learn what to forget and retain. Complete episodes are replayed through the GRU so late losses can alter earlier encoding: backpropagation through time. A slowly moving target network stabilizes TD targets, following deep Q-learning practice [11]:

$$\theta^- \leftarrow (1 - \tau)\theta^- + \tau\theta. \quad (15)$$

### 3.5 Evaluation principles

RL evaluation is difficult because training changes the data distribution and results vary across seeds. Reproducibility studies warn that few-run point estimates can reverse rankings [16, 17]. Evaluation should separate exploratory training from held-out greedy behavior, show per-task and aggregate results, and distinguish success probability from speed. Conditional duration can reward a reckless policy because failures are excluded. We therefore treat success as primary, label conditional steps, and retain full finite-horizon success curves. With two seeds, this study is exploratory rather than inferential.

## 4 Sequence goals and three geometries

### 4.1 Goal-conditioned control

A goal is an ordered sequence  $g = ((a_1^g, o_1^g), \dots, (a_m^g, o_m^g))$ . Other events may occur between checkpoints. Progress  $j_t$  is allowed because it is computed from the experienced history. Rewards are 10 for completion, 1 for an intermediate checkpoint, and  $-0.01$  otherwise. A goal-conditioned value  $Q(z, g, j, a)$  lets the same remembered history support different policies.

## 4.2 Symmetric strategic similarity

Each known goal has embedding  $e_g \in \mathbb{R}^k$ . Distance  $D_E(g, h) = \|e_g - e_h\|_2$  is intended to mean similarity of strategy, not labels. At encountered state  $z$ , define

$$\pi_g(a|z, j) = \text{softmax}_a(Q(z, g, j, a)/T). \quad (16)$$

For policies  $p, q$ , Jensen–Shannon divergence is

$$D_{\text{JS}}(p, q) = \frac{1}{2}D_{\text{KL}}(p\|m) + \frac{1}{2}D_{\text{KL}}(q\|m), \quad m = \frac{1}{2}(p + q). \quad (17)$$

Its square root is a metric [13]. The regularizer is

$$\mathcal{L}_{\text{geom}} = \frac{1}{|\mathcal{P}|} \sum_{g < h} [\|e_g - e_h\|_2 - c\sqrt{D_{\text{JS}}(\pi_g, \pi_h)}]^2. \quad (18)$$

The policy target is detached. Geometry is therefore partly imposed. Held-out agreement tests generalization; it is not spontaneous emergence without bias.

## 4.3 Trajectory similarity

Local policy similarity may miss long-horizon differences. From held-out rollouts we build

$$\phi(g) = [\text{action frequencies, action/outcome frequencies, normalized duration, success rate}] \quad (19)$$

and  $D_{\text{traj}}(g, h) = \|\phi(g) - \phi(h)\|_2$ . This resembles successor features, which describe policies by expected future features and support transfer [14], although our signature is diagnostic rather than a learned successor model.

## 4.4 Directed reachability and displacement

Let  $\tau_g$  be first completion time. The ideal intervention-count distance is

$$d^*(z, g, j) = \min_{\pi} \mathbb{E}_{\pi}[\tau_g|z, j]. \quad (20)$$

This is a stochastic shortest-path problem [12] with Bellman form

$$d^*(z, g, j) = 1 + \min_a \sum_o P(o|z, a) d^*(F(z, a, o), g, j'(a, o)). \quad (21)$$

The network learns positive action cost  $C(z, g, j, a)$  and sets  $d(z, g, j) = \min_a C(z, g, j, a)$ . Its target is 1 on completion and otherwise

$$y_t^C = 1 + \min_{a'} C_{\theta^-}(z_{t+1}, g, j_{t+1}, a'). \quad (22)$$

This is directed: measurement can destroy conditions needed to return.

For  $N$  goals, the agent’s goal-relative position is

$$\mathbf{D}(z) = (d(z, g_1), \dots, d(z, g_N)). \quad (23)$$

An observed intervention produces displacement  $\Delta \mathbf{D}_t = \mathbf{D}(z_{t+1}) - \mathbf{D}(z_t)$ . Negative coordinates mean closer, positive farther. With the learned predictor, a contemplated action has

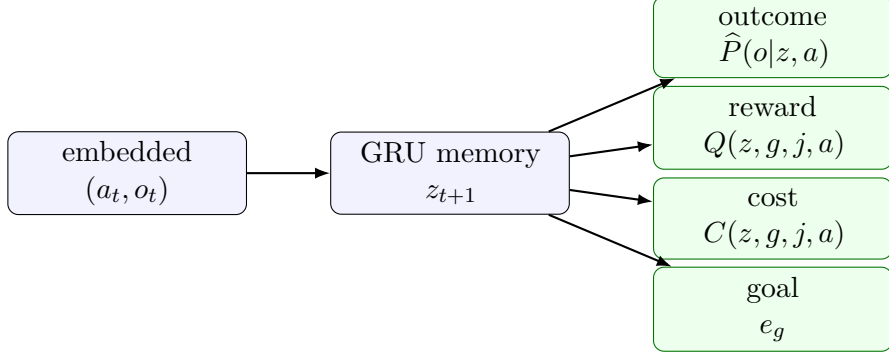
$$\mathbb{E}[d'_g|z, a] = \sum_o \hat{P}(o|z, a) d(F(z, a, o), g), \quad (24)$$

which provides a route to model-based planning.

A scalar mean hides risk, so we also estimate

$$R_g(z, T) = \mathbb{P}(\tau_g \leq T|z, \pi_g). \quad (25)$$

This parallels distributional RL’s emphasis on distributions rather than only expected return [15].



**Figure 2:** Shared recurrent representation and four learning objectives. Outcome prediction is goal-independent; reward and cost are goal-conditioned.

**Table 2:** Production configuration.

|                                   |             |                          |           |
|-----------------------------------|-------------|--------------------------|-----------|
| Training episodes/run             | 400         | Evaluation episodes/goal | 100       |
| Geometry episodes/goal            | 40          | Horizon                  | 15        |
| Discount $\gamma$                 | 0.95        | Tabular $\alpha$         | 0.10      |
| History length                    | 4           | Neural learning rate     | $10^{-3}$ |
| $\epsilon$ start/end/decay        | 1/.05/.99   | Target $\tau$            | .02       |
| GRU/goal dimensions               | 32/6        | Interaction embedding    | 8         |
| $\lambda_P, \lambda_C, \lambda_G$ | .5, .5, .05 | Geometry scale           | 2         |
| Weak strength                     | .8          | Seeds                    | 0,1       |

## 5 Architecture and experimental protocol

All agents share one interaction protocol and receive only goal ID, progress, action, outcome, reward, and termination. The **tabular** state is [equation \(10\)](#) with  $L = 4$ . The **GRU** concatenates memory, a learned goal embedding, and one-hot progress before an MLP Q-head. The **predictive geometry GRU** adds a goal-independent outcome head, goal-conditioned cost head, and [equation \(18\)](#). Its prediction loss is

$$\mathcal{L}_{\text{pred}} = -\log \hat{P}_\theta(o_t|z_t, a_t), \quad (26)$$

computed before incorporating  $o_t$ . The full loss is

$$\mathcal{L} = \mathcal{L}_Q + \lambda_P \mathcal{L}_{\text{pred}} + \lambda_C \mathcal{L}_C + \lambda_G \mathcal{L}_{\text{geom}}. \quad (27)$$

Goals are sampled uniformly. Each episode resets hidden state, memory, and progress. Trajectories are replayed once for backpropagation through time; reward and cost use [equation \(15\)](#); gradients are clipped to norm 5.

Evaluation disables exploration. Separate geometry rollouts evaluate every goal-conditioned policy at each state, build trajectory signatures, compare blank-history cost to hitting time, estimate [equation \(25\)](#), and group displacements by action/outcome. PCA is used only to visualize embeddings; all distance comparisons use original dimensions.

The study contains 14,400 training episodes (107,804 interventions), 26,400 evaluation episodes (179,342 interventions), and 3,520 geometry episodes. An exact manifest, tidy CSVs, model weights, and plots are saved. Eighteen tests verify quantum validity, outcome masks, reproducibility, goals, runner semantics, every agent, and geometry utilities.



**Table 3:** Success rate, mean  $\pm$  sample standard deviation over two seeds. Bold is the largest scenario mean.

| World / state            | Finite history                    | GRU                               | Predictive geometry GRU           |
|--------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Z/X/weak-Z / $ 1\rangle$ | .810 $\pm$ .012                   | <b>.874 <math>\pm</math> .038</b> | .861 $\pm$ .035                   |
| Z/X/weak-Z / $ +\rangle$ | .878 $\pm$ .009                   | .809 $\pm$ .063                   | <b>.924 <math>\pm</math> .005</b> |
| Pauli / $ +i\rangle$     | <b>.863 <math>\pm</math> .018</b> | .806 $\pm$ .057                   | .825 $\pm$ .169                   |
| unsharp / $I/2$          | <b>.849 <math>\pm</math> .008</b> | .788 $\pm$ .080                   | .820 $\pm$ .071                   |
| Pauli+SIC / $I/2$        | <b>.713 <math>\pm</math> .006</b> | .649 $\pm$ .027                   | .697 $\pm$ .172                   |
| qutrit MUB / $ 2\rangle$ | .634 $\pm$ .034                   | <b>.734 <math>\pm</math> .082</b> | .703 $\pm$ .015                   |
| Macro-average            | .791                              | .777                              | <b>.805</b>                       |

## 6 Results

### 6.1 Control performance

No backend dominates. Recurrent memory helps on the qutrit and original  $|1\rangle$  scenarios; tabular learning leads on Pauli, unsharp, and Pauli+SIC; the full model is strongest from  $|+\rangle$ . Macro conditional steps are 4.62 (tabular), 4.55 (GRU), and 4.41 (full), but success remains primary.

Neural instability is substantial. Mean absolute seed gaps are 2.1 percentage points for tabular, 8.2 for GRU, and 11.0 for the full model; the largest full gap is 24.3. Both recurrent agents have zero success on the three-checkpoint qutrit goal, while tabular reaches 15%. The plain GRU also has zero success on the two-step  $Z0\_SIC0$  goal.

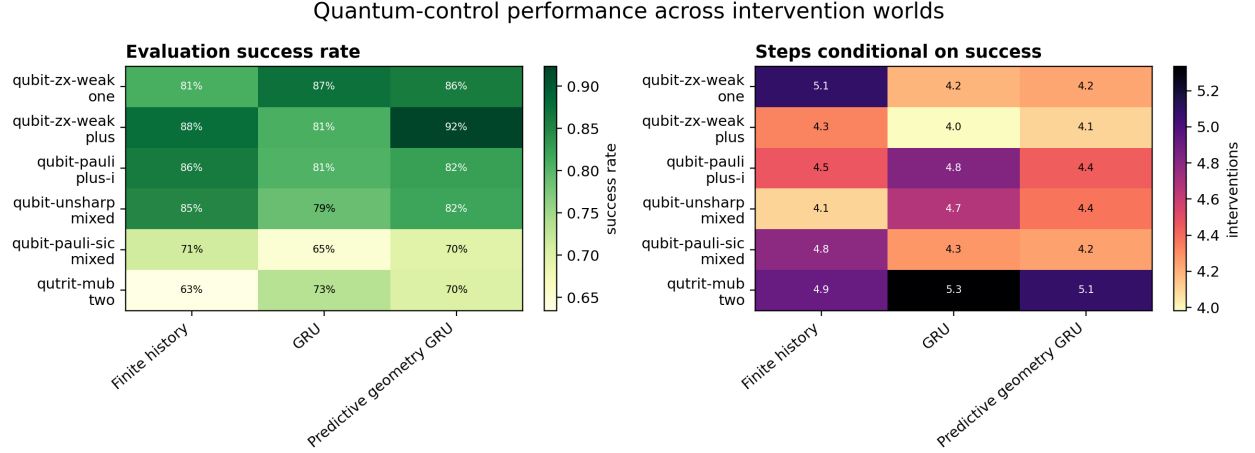
### 6.2 Goal-geometry validation

Across 12 full-agent runs, embedding distance has mean Spearman correlation 0.537 with held-out policy strategy distance (median 0.533, standard deviation 0.207, range 0.121–0.825). This is meaningful but imperfect generalization of the regularized relation. Correlation with more independent trajectory distance is weaker: mean 0.311, median 0.274, standard deviation 0.303, range  $-0.091$ – $0.786$ . Geometry is not yet a unique robust object. The first two PCA axes explain 77.3% of embedding variance on average.

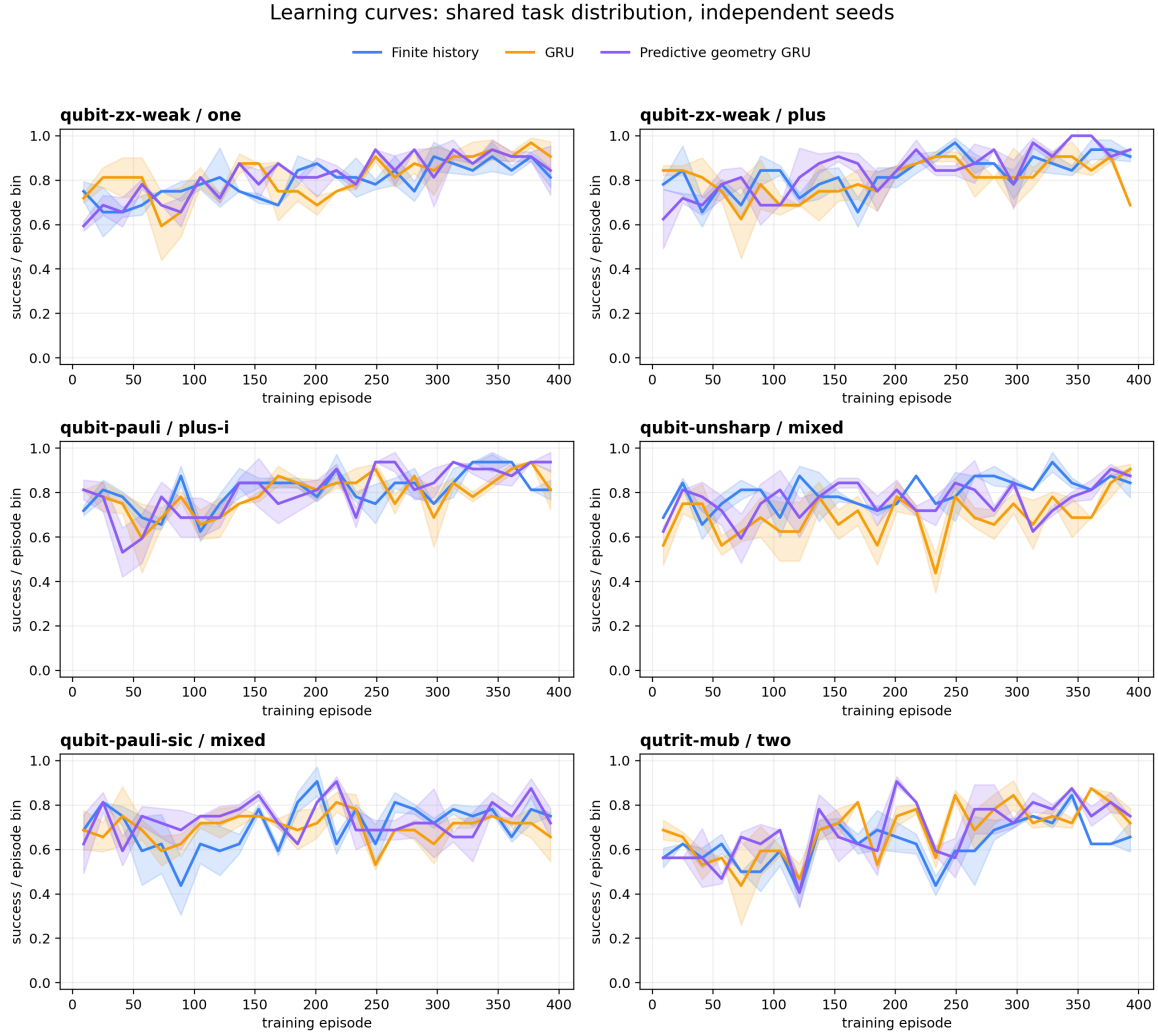
### 6.3 Reachability and intervention displacement

Across 79 goals with at least one geometry-rollout success, predicted cost correlates  $r = 0.471$  with conditional empirical steps and has mean absolute error 1.75. Predictions are compressed to 2.46–5.89 while observed means span 1.0–10.71. The head contains ordinal information but is not a calibrated clock. Off-policy minimum backups, function approximation, policy mismatch, finite-horizon censoring, and conditioning on success all contribute.

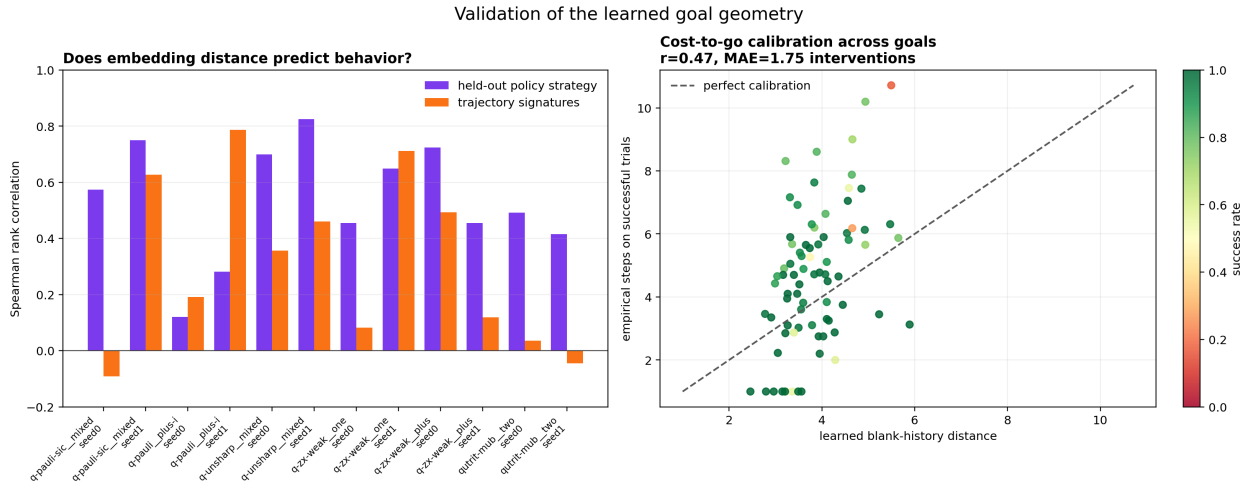
In [figure 8](#),  $Z:0$ ,  $X:0$ , and  $Y:0$  change matching-goal distance by  $-1.58$ ,  $-1.80$ , and  $-1.91$ . The four SIC outcomes change their matching distances by  $-2.43$  to  $-2.95$ , while most nonmatching coordinates increase. Completion explicitly sets a matching goal to zero, so diagonals should not be overinterpreted; structured off-diagonal change is the more interesting signature of backaction.



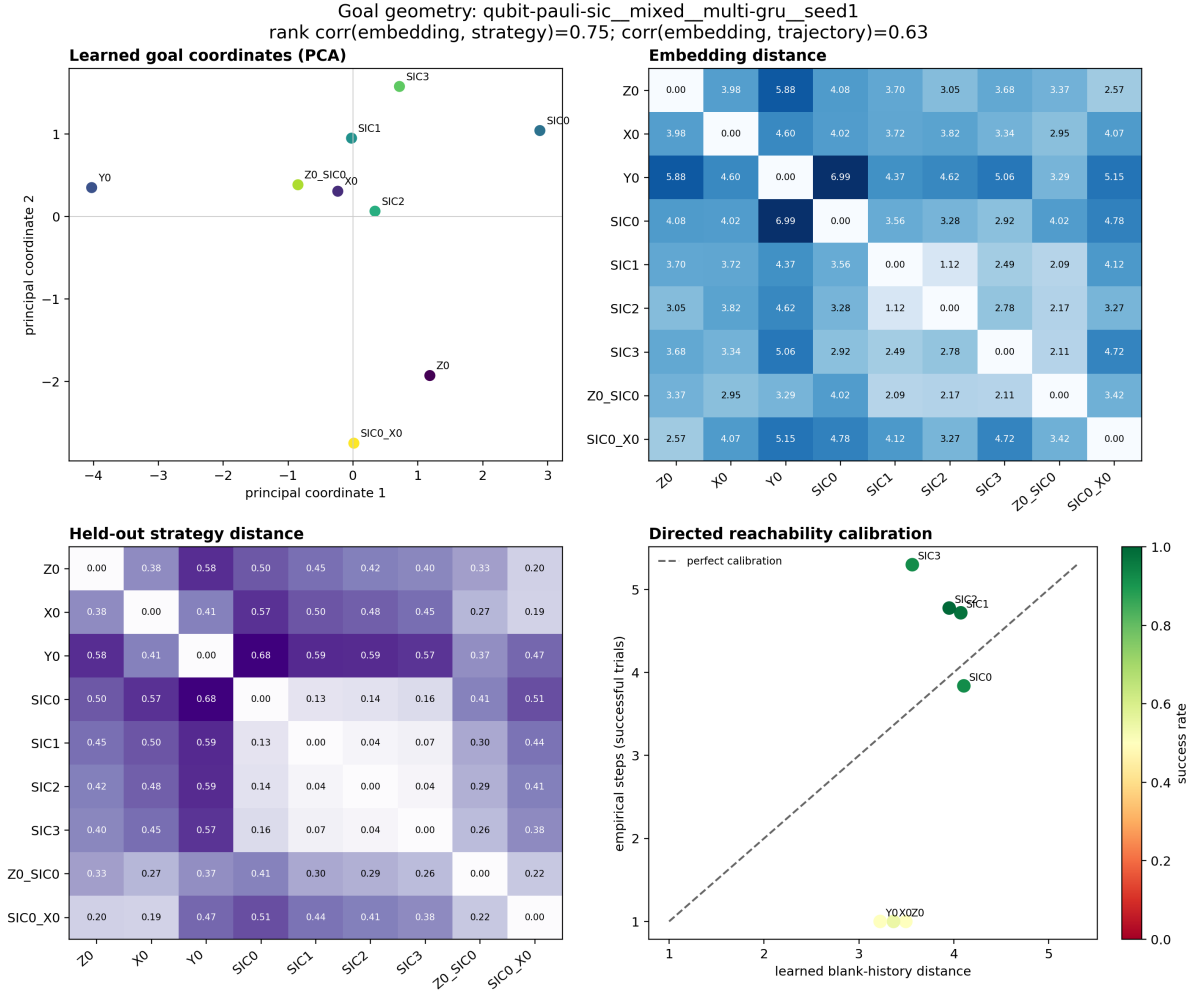
**Figure 3:** Held-out success and conditional intervention count, averaged over two seeds. Conditional steps exclude failures and must be read beside success.



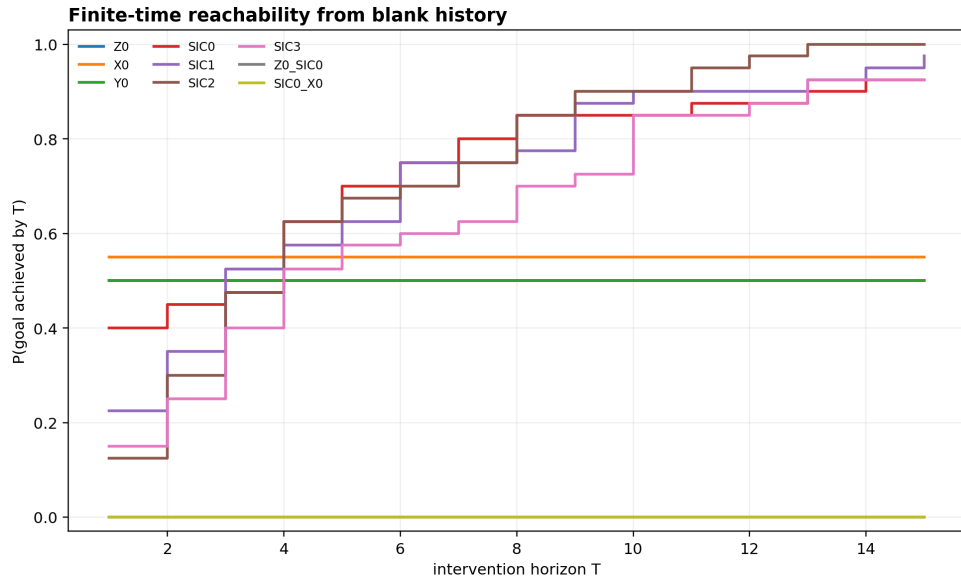
**Figure 4:** Binned exploratory training success. Shading is standard error over only two seeds and is descriptive, not a reliable interval.



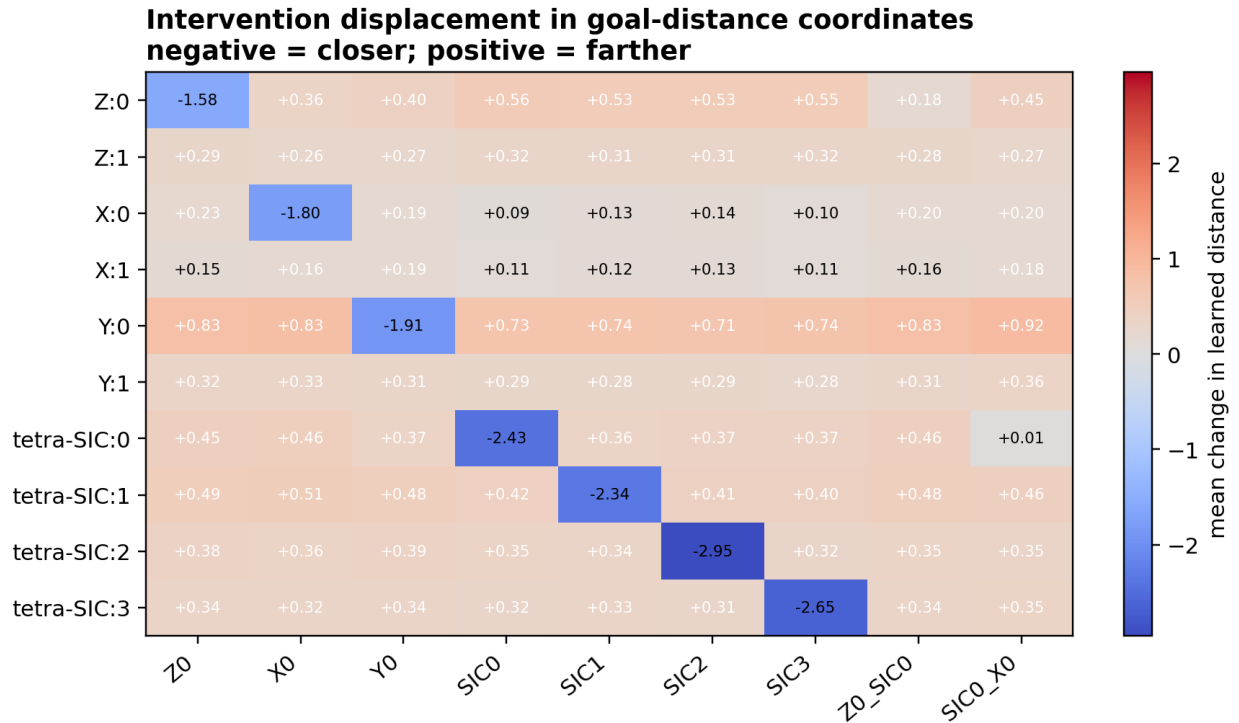
**Figure 5:** Left: correlation of embedding distance with held-out policy and trajectory distances. Right: cost-to-go calibration across successful goals.



**Figure 6:** Representative Pauli+SIC geometry (seed 1): PCA coordinates, full embedding distances, held-out policy distances, and reachability calibration. The embedding has strategy correlation 0.75 and trajectory correlation 0.63 in this run.



**Figure 7:** Finite-time success curves retain reliability information hidden by a conditional mean hitting time.



**Figure 8:** Mean displacement in learned goal-distance coordinates. Blue means closer and red farther. Matching outcomes create strong negative diagonals, while backaction alters multiple alternative affordances.

## 7 Inverse-designing emergent two-dimensional space

### 7.1 Hodological criterion and bilevel objective

The preceding experiments ask whether a finite goal repertoire acquires useful structure. We now ask the inverse question: can one choose a quantum world and goals such that difficulty itself is compatible with ordinary 2D distance? For operational states  $s_i$  associated with goals  $g_i$ , let

$$D_{ij} = \mathbb{E}_{\pi_j}[\tau_{g_j} \mid s_i] - c_{\text{probe}}, \quad (28)$$

where the common terminal-probe cost is removed. A Euclidean interpretation in  $d$  dimensions is tested by metric multidimensional scaling (MDS), minimizing Kruskal-style normalized raw stress [22]:

$$\text{Stress}_d(D, X) = \sqrt{\frac{\sum_{i < j} (\|x_i - x_j\| - D_{ij})^2}{\sum_{i < j} D_{ij}^2}}. \quad (29)$$

Classical MDS initializes deterministic SMACOF majorization. Manifold learning similarly reconstructs global geometry from local metric information [23], but here the dissimilarities are learned control costs.

Let  $\theta$  parameterize the state and instruments and  $\mathcal{G}$  the goals. The intended general bilevel design is

$$(\theta^*, \mathcal{G}^*) = \arg \min_{\theta, \mathcal{G}} [\lambda_s \text{Stress}_d + \lambda_c P_{\text{calibration}} + \lambda_n P_{\text{nonmetric}} + \lambda_a P_{\text{asymmetry}} + \lambda_f P_{\text{failure}} + \lambda_k P_{\text{complexity}}], \quad (30)$$

where the matrix inside the objective is produced by an inner RL learner. The first experiment simplifies this expensive problem: its exact stochastic shortest paths are known to the privileged outer loop, which searches one instrument parameter before independent policies are trained and evaluated.

### 7.2 Nine-dimensional quantum construction

Let

$$\mathcal{H}_9 = \text{span}\{|x, y\rangle : x, y \in \{0, 1, 2\}\}, \quad \rho_0 = |1, 1\rangle\langle 1, 1|. \quad (31)$$

The labels  $(x, y)$  are private validation metadata; the agent receives only an arbitrary place symbol. Four axial and four diagonal actions map a site  $s$  to an adjacent site  $f_a(s)$ . Their hidden source-resolved Kraus events are

$$K_{s,+}^{(a)} = \sqrt{p_a(s)} |f_a(s)\rangle\langle s|, \quad K_{s,-}^{(a)} = \sqrt{1 - p_a(s)} |s\rangle\langle s|. \quad (32)$$

At open boundaries  $p_a(s) = 0$ , so summing equation (32) over sources and branches gives  $I$ . Axial moves have probability one; legal diagonals share  $p_d$ . Several unobserved source events are coarse-grained into one observed destination symbol. The corresponding instrument operation is

$$\mathcal{I}_y^{(a)}(\rho) = \sum_{s: f_a(s)=y} K_{s,+}^{(a)} \rho K_{s,+}^{(a)\dagger} + K_{s,-}^{(a)} \rho K_{s,-}^{(a)\dagger}. \quad (33)$$

A ninth, common place probe has projectors  $P_s = |s\rangle\langle s|$ . Goal  $g_s$  is simply “perform this probe and obtain symbol  $s$ .” No goal-specific destination action exists.

Training uniformly prepares all nine localized sources and supplies one place probe before task time. This contextual outcome localizes the agent but gives no adjacency, axis, orientation, or

**Table 4:** Learned spatial geometry, mean  $\pm$  sample standard deviation over three seeds.  $R_P^2$  aligns MDS coordinates with concealed coordinates only after training.

| Metric                 | Optimized observed                | Optimized blind | Cardinal observed |
|------------------------|-----------------------------------|-----------------|-------------------|
| All-pairs success      | 1.000 $\pm$ .000                  | .482 $\pm$ .023 | 1.000 $\pm$ .000  |
| 1D stress              | .372 $\pm$ .011                   | .450 $\pm$ .042 | .460 $\pm$ .000   |
| 2D stress              | <b>.071 <math>\pm</math> .011</b> | .233 $\pm$ .032 | .142 $\pm$ .000   |
| 3D stress              | .064 $\pm$ .011                   | .228 $\pm$ .038 | .141 $\pm$ .000   |
| Procrustes $R_P^2$     | .975 $\pm$ .005                   | .965 $\pm$ .011 | 1.000 $\pm$ .000  |
| Exact-cost correlation | .936 $\pm$ .003                   | .865 $\pm$ .010 | 1.000 $\pm$ .000  |
| Directionality         | .122 $\pm$ .037                   | .264 $\pm$ .093 | .000 $\pm$ .000   |

metric. Its Markov state is  $(g, p, \ell)$ : goal ID, progress, and latest arbitrary place symbol. Ordinary tabular Q-learning then discovers the transitions. Two controls separate possible explanations: a *blind* instrument reports only movement success/failure, and a place-observed *cardinal* world removes diagonals, leaving Manhattan shortest paths.

### 7.3 Outer optimization and learned geometry

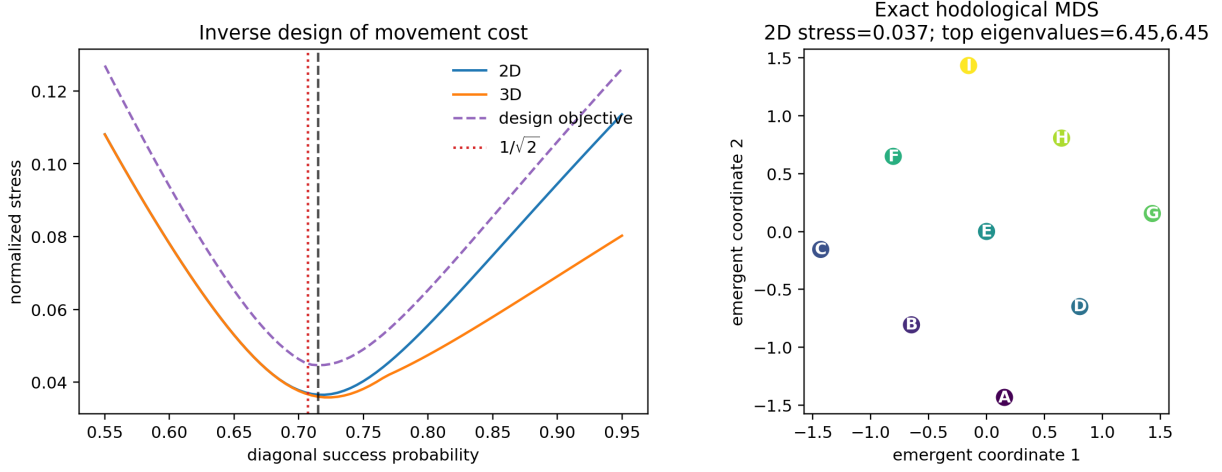
One axial displacement costs one intervention; repeated diagonal attempts cost  $1/p_d$  on average. Euclidean cost matching suggests  $p_d \simeq 1/\sqrt{2}$ . Searching 81 values in  $[0.55, 0.95]$  selects  $p_d = 0.715$  by a combination of 2D stress, negative centered-Gram spectrum, and privileged coordinate recovery. The proximity to  $1/\sqrt{2}$  is an outcome of the search, not an input to its objective.

The selected exact matrix has 1D/2D/3D stresses 0.4305/0.0365/0.0360, contains 93.4% of its positive Gram-spectrum variance in two dimensions, and correlates 0.995 with concealed Euclidean distance. The cardinal control has 2D stress 0.1416 and negative-spectrum fraction 0.217, versus 0.081 for the selected instrument. Thus a grid-like arrangement does not by itself make the operational metric Euclidean.

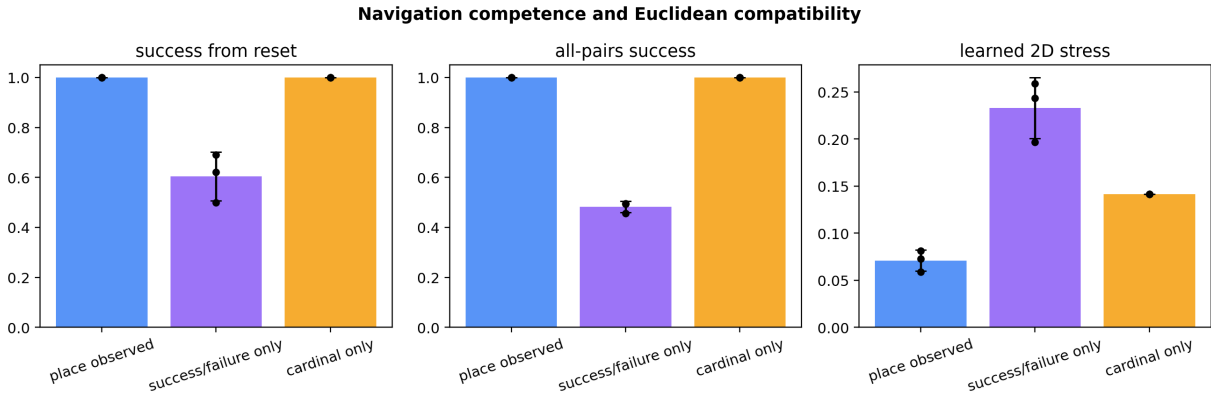
Each of three worlds and three seeds receives 6,000 uniformly sampled source-goal episodes. Evaluation uses 100 trials for all 81 ordered pairs: 54,000 training and 72,900 evaluation episodes in total. Failures contribute a finite restricted-mean penalty at horizon 12. The learned matrix is symmetrized for MDS, while normalized antisymmetry is retained separately.

The proposed construction has a decisive dimension gap: learned stress falls from 0.372 in 1D to 0.071 in 2D, while a third coordinate improves it only to 0.064. Procrustes  $R^2 = 0.975$  confirms recovery of the concealed arrangement up to rotation, reflection, translation, and scale. Coordinate recovery alone would be misleading: the cardinal control has  $R_P^2 \simeq 1$  but twice the 2D stress because its metric is Manhattan.

The negative stages are informative. Center-only training achieved 0.92–0.95 reset success but only about 0.25 arbitrary all-pairs success: a star of routes from one origin is not a space. Random-source binary observation reached only 0.482 all-pairs success. Destination observation with a six-event literal history achieved nearly perfect control but split one place by arrival route and left 2D stress near 0.16. Quotienting histories by their latest place symbol yields the final result.

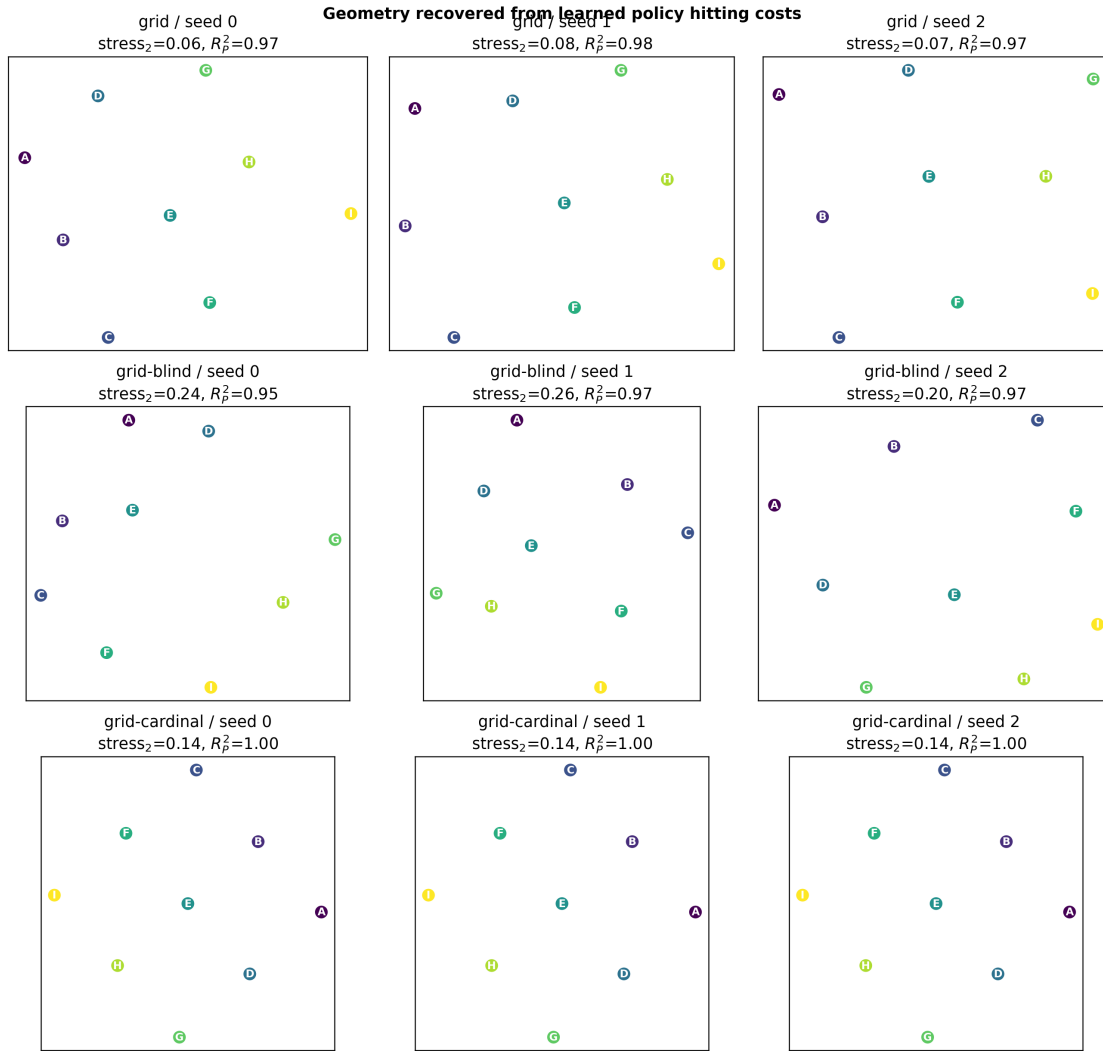


**Figure 9:** Inverse design of diagonal success probability (left) and exact MDS of the selected hitting-cost matrix (right). The red line is  $1/\sqrt{2}$ ; the black line is the selected 0.715. Exact 2D stress is 0.0365. Place labels carry no coordinate information.

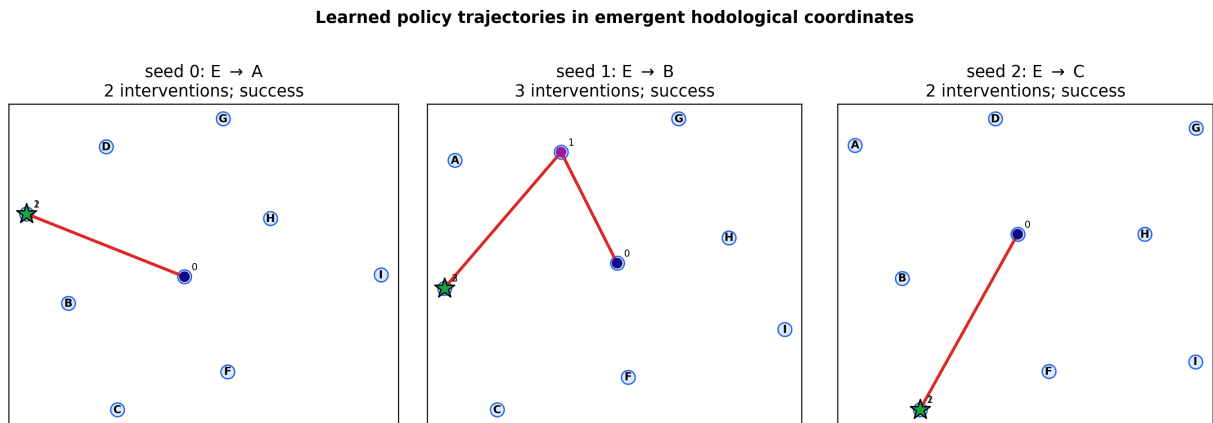


**Figure 10:** Navigation and Euclidean compatibility. Destination-place observations make the optimized world fully navigable and halve 2D stress relative to the equally competent cardinal/Manhattan control. Binary success/failure observation damages both competence and geometry.





**Figure 11:** MDS of empirical policy hitting costs. Rows show the proposed world, the partial-observability control, and the cardinal control. Axes have no externally meaningful orientation.



**Figure 12:** Observed place sequences under learned policies, drawn in coordinates computed only from learned all-pairs difficulty. The green star is the goal; numbers mark intervention order. In this limited sense, action histories are stochastic motion through emergent hodological space.

## 8 Discussion and limitations

Useful measurement strategies can be learned without exposing hidden quantum variables. Recurrent memory sometimes helps, but a short literal history is a strong small-world baseline. Goal embeddings carry moderate behavioral meaning, although local policy geometry transfers more reliably than whole-trajectory geometry. Directed coordinates expose coherent measurement displacements even though scalar cost is poorly calibrated.

The spatial study gives a stronger but narrower positive statement: there exists a valid quantum instrument/goal construction in which policy-derived difficulty selects two dimensions, recovers a concealed lattice, and turns observed action sequences into spatial paths. Its controls show that hidden connectivity, coordinate recovery, control success, and Euclidean metric fit are distinct requirements.

These results do not show that a Bloch sphere, Hilbert space, or uniquely quantum geometry spontaneously emerged. Such a claim would require latent-state analysis, intervention ablations, classical controls, and privileged offline decoding of quantum invariants. Other limitations are:

- two seeds cannot support significance claims; modern RL methodology recommends more runs, intervals, robust aggregates, and performance profiles [17];
- episode budgets are equal, not compute budgets, and hyperparameters are not independently optimized;
- horizon 15 censors hard goals, reward shaping affects behavior, and conditional steps omit failures;
- one-step prediction does not guarantee predictive sufficiency;
- goal embeddings are lookup entries and cannot represent unseen goals;
- embedding targets come from the same Q-head, creating shared-error and circularity risks despite held-out evaluation;
- cost is learned off-policy from reward-oriented behavior and is not constrained to satisfy stochastic-shortest-path consistency;
- simulators omit Hamiltonian dynamics, decoherence, detector error, latency, and hardware drift.
- the spatial construction uses localized qudit states and entanglement-breaking measure-and-prepare channels; latent places, open boundaries, and a sharp localization probe are designed into the simulator;
- the  $3 \times 3$  lattice is too small to establish continuum structure, topology recovery, scale dependence, or curvature, and three spatial seeds remain exploratory;
- symmetrizing learned hitting costs is appropriate for testing a Euclidean hypothesis but discards directed information that a more general hodological geometry must retain.

## 9 Outlook

**Compositional goals and transfer.** Encode checkpoint sequences with a GRU or transformer, hold out complete goals, and test zero/few-shot transfer. Geometry matters if proximity predicts adaptation speed. Successor features provide a principled comparison [14].

**Multi-step predictive state.** Train action-conditional predictions over entire future tests, compare recurrent and transformer memory, and measure predictive calibration by horizon. Vary informational completeness to determine which latent dimensions disappear.

**Planning in possibility space.** Use [equation \(24\)](#) to compare direct Q-control, greedy distance reduction, and tree search through imagined outcomes. The learned geometry should improve decisions, not only visualization.

**Distributional reachability.** Learn  $\tau_g$  distributions or  $R_g(z, T)$  directly, using proper scoring rules and reliability plots. Distributional Bellman methods are a starting point [\[15\]](#).

**Directed non-Euclidean structure.** Explore directed graphs, asymmetric energy models, quasimetrics, hyperbolic goal hierarchies, and Finsler-like local costs. Choose geometry through predictive validity and planning utility, not aesthetics.

**Optimize states, instruments, and landmarks.** Generalize [equation \(30\)](#) from one success probability to positive Choi matrices or Stinespring isometries, enforcing complete positivity and trace preservation. Jointly select goal landmarks for coverage and stable trilateration, penalizing nonlocal teleportation and goal-specific actions. Evolutionary search is a robust first outer loop; differentiable meta-learning becomes attractive once the inner planner is model-based.

**Remove explicit place localization.** Alias, weaken, or noise destination outcomes and require a recurrent predictive state to recover location. Scaling from grids to tori, cylinders, spheres, and defected graphs should be evaluated with out-of-sample stress, local dimension, neighborhood preservation, and topology, not only global MDS [\[23\]](#).

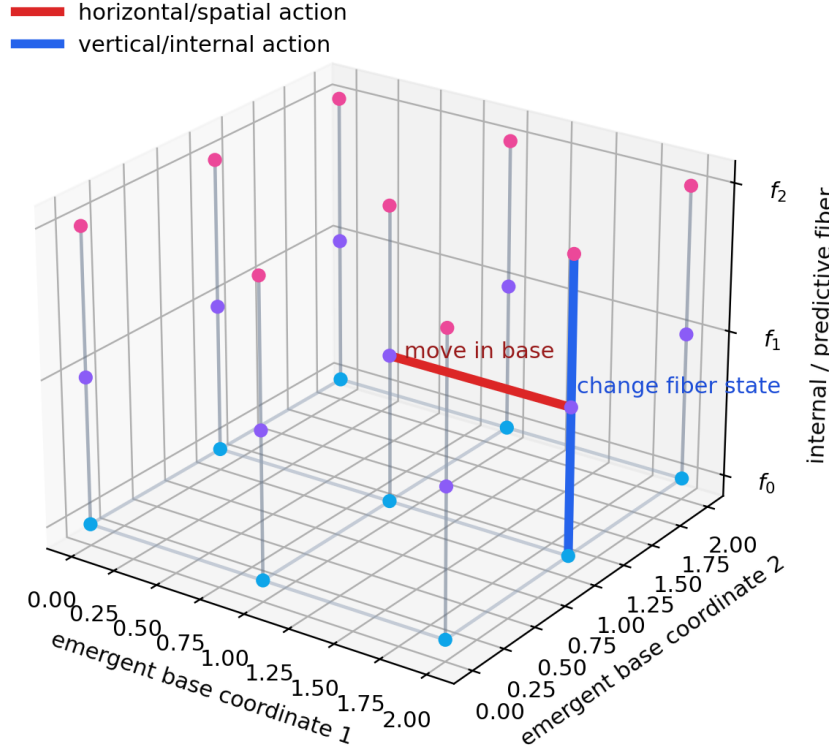
**Spatial base and internal fiber.** Give each place several internal quantum states and goals, then test whether the full hodological state space  $E$  locally decomposes as  $F_b \hookrightarrow E \xrightarrow{\pi} B$ , with a stable 2D or 3D spatial base  $B$  and internal fiber  $F_b$  [\[24\]](#). Actions may be horizontal, vertical, or coupled. Learned transition maps between local charts and the failure of an internal state to return after transport around a closed base loop provide operational notions of connection and holonomy.

**Curvature and spacetime.** Position-dependent transition kernels permit local metric and transport curvature tests; Ollivier curvature supplies one principled discrete diagnostic for Markov processes [\[25\]](#). This is only a bridge. General relativity relates spacetime curvature to dynamics [\[26\]](#), so a credible analogue would require robust local geometry, connection, sources, and an independently tested field relation. Time cannot simply be appended as a Euclidean coordinate: future-goal reachability must first yield causal order, directed cones, clock tasks, and observer-dependent simultaneity.

**Information-seeking action.** Replace undirected epsilon exploration with expected information gain, prediction uncertainty reduction, or reachability uncertainty reduction. This turns experimentation itself into goal-directed behavior.

**Privileged offline physics.** Pair learned  $z_t$  with hidden Bloch coordinates only after training. Test decodability, intrinsic dimension, convexity, and equivalence classes; compare against classical partially observed environments.

Fiber-bundle research target (schematic, not an empirical result)



**Figure 13:** Proposed spatial-base/internal-fiber program. This is a schematic, not an empirical result. Horizontal actions move through emergent place; vertical actions alter internal possibilities.

**Stronger statistics and hardware.** Use 10–20 seeds, longer budgets, bootstrap intervals, interquartile means, and performance profiles [16, 17]. Then add decoherence, readout error, continuous weak measurement, and device drift, ultimately learning from laboratory records as envisioned in model-free quantum control [19].

## 10 Conclusion

An agent beginning only with interventions and outcomes can learn more than a policy. It can compress history, predict contemplated measurements, compare aims by behavior, estimate goal-relative reachability, and represent measurement backaction as movement through future possibility. The present study makes that program concrete but also establishes a demanding standard: an attractive embedding is not self-validating. It must predict independent behavior, transfer, and reachability.

The inverse-designed qudit world additionally establishes the first spatial existence result: all-pairs learned difficulty robustly prefers two dimensions and learned policies trace paths in that emergent map. Just as importantly, the failures show that a single-origin task, a hidden grid, or a visually grid-like embedding is not enough. Broad source coverage, operational localization or memory, and approximately isotropic movement costs are jointly required.

The geometry-aware agent is competitive but not universally superior. Its embedding has measurable, seed-dependent strategic meaning; its directed cost orders some goals but is not accurate in absolute units. These shortcomings identify the next experiments. The long-term prospect is an agent not handed a physical state space, but able to construct a predictive present and a geometry of possible aims from action and consequence alone.

## A Reproduction and artifacts

```
conda run -n qbist_spacetime python run_experiment_suite.py \
  --scenarios qubit-zx-weak:one,qubit-zx-weak:plus,\
qubit-pauli:plus-i,qubit-unsharp:mixed,\
qubit-pauli-sic:mixed,qutrit-mub:two \
  --backends tabular,gru,multi-gru --seeds 0,1 \
  --episodes 400 --eval-episodes 100 --geometry-episodes 40 \
  --max-steps 15 --epsilon-decay 0.99 --device cpu \
  --output results/comparative
python plot_results.py results/comparative

python spatial_hodology.py --output results/spatial-hodology \
  --seeds 0,1,2 --episodes 6000 --pair-episodes 100 \
  --max-steps 12
```

The manifest records exact runs; episode CSVs retain raw outcomes; geometry directories store embeddings, three distance views, reachability, curves, and displacements; and the plot tree contains 51 figures. The spatial bundle separately retains its 81-point instrument search, exact and learned all-pairs matrices, text Q-tables, trajectory records, manifest, and five figures.

## B Goal repertoires

### qubit-zx-weak

$Z0, Z1, X0, X1, Z0\_X0, X0\_Z0, weakZ0\_Z0.$

### qubit-pauli

$Z0, X0, Y0, Z0\_X0, X0\_Z0, Y0\_X1, Z0\_X0\_Y0.$

### qubit-unsharp

$wZ0, wX0, wY0, wZ0\_wX0, wX0\_wZ0, wZ0\_wX0\_wY0.$

### **qubit-pauli-sic**

$Z0, X0, Y0, SIC0, SIC1, SIC2, SIC3, Z0\_SIC0, SIC0\_X0.$

### **qutrit-mub**

$Z0, Z1, Z2, F0, F1, Z0\_F0, F0\_Z0, Z0\_F0\_P0.$

### **spatial worlds**

$A, B, C, D, E, F, G, H, I$ , each denoting one outcome of the same common place probe.

## **References**

- [1] C. A. Fuchs, N. D. Mermin, and R. Schack, “An introduction to QBism with an application to the locality of quantum mechanics,” *American Journal of Physics* 82, 749–754 (2014). [doi:10.1119/1.4874855](https://doi.org/10.1119/1.4874855).
- [2] K. Kraus, *States, Effects, and Operations* (Springer, 1983). [doi:10.1007/3-540-12732-1](https://doi.org/10.1007/3-540-12732-1).
- [3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed. (Cambridge University Press, 2010). [doi:10.1017/CBO9780511976667](https://doi.org/10.1017/CBO9780511976667).
- [4] J. M. Renes, R. Blume-Kohout, A. J. Scott, and C. M. Caves, “Symmetric informationally complete quantum measurements,” *Journal of Mathematical Physics* 45, 2171–2180 (2004). [doi:10.1063/1.1737053](https://doi.org/10.1063/1.1737053).
- [5] T. Durt, B.-G. Englert, I. Bengtsson, and K. Życzkowski, “On mutually unbiased bases,” *International Journal of Quantum Information* 8, 535–640 (2010). [doi:10.1142/S0219749910006502](https://doi.org/10.1142/S0219749910006502).
- [6] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. (MIT Press, 2018). <https://mitpress.mit.edu/9780262039246/reinforcement-learning/>.
- [7] C. J. C. H. Watkins and P. Dayan, “Q-learning,” *Machine Learning* 8, 279–292 (1992). [doi:10.1007/BF00992698](https://doi.org/10.1007/BF00992698).
- [8] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, “Planning and acting in partially observable stochastic domains,” *Artificial Intelligence* 101, 99–134 (1998). [doi:10.1016/S0004-3702\(98\)00023-X](https://doi.org/10.1016/S0004-3702(98)00023-X).
- [9] M. L. Littman, R. S. Sutton, and S. Singh, “Predictive representations of state,” in *NeurIPS 14* (2001). <https://papers.nips.cc/paper/2001/hash/1e4d36177d71bbb3558e43af9577d70e-Abstract.html>.
- [10] K. Cho et al., “Learning phrase representations using RNN encoder–decoder,” in *EMNLP*, 1724–1734 (2014). [doi:10.3115/v1/D14-1179](https://doi.org/10.3115/v1/D14-1179).
- [11] V. Mnih et al., “Human-level control through deep reinforcement learning,” *Nature* 518, 529–533 (2015). [doi:10.1038/nature14236](https://doi.org/10.1038/nature14236).
- [12] D. P. Bertsekas and J. N. Tsitsiklis, “An analysis of stochastic shortest path problems,” *Mathematics of Operations Research* 16, 580–595 (1991). [doi:10.1287/moor.16.3.580](https://doi.org/10.1287/moor.16.3.580).
- [13] D. M. Endres and J. E. Schindelin, “A new metric for probability distributions,” *IEEE Transactions on Information Theory* 49, 1858–1860 (2003). [doi:10.1109/TIT.2003.813506](https://doi.org/10.1109/TIT.2003.813506).

- [14] A. Barreto et al., “Successor features for transfer in reinforcement learning,” in *NeurIPS 30* (2017). <https://papers.nips.cc/paper/6994-successor-features-for-transfer-in-reinforcement-learning>.
- [15] M. G. Bellemare, W. Dabney, and R. Munos, “A distributional perspective on reinforcement learning,” *Proceedings of Machine Learning Research* 70, 449–458 (2017). <https://proceedings.mlr.press/v70/bellemare17a.html>.
- [16] P. Henderson et al., “Deep reinforcement learning that matters,” in *AAAI* (2018). doi:10.1609/aaai.v32i1.11694.
- [17] R. Agarwal et al., “Deep reinforcement learning at the edge of the statistical precipice,” in *NeurIPS 34*, 29304–29320 (2021). <https://papers.nips.cc/paper/2021/hash/f514cec81cb148559cf475e7426eed5e-Abstract.html>.
- [18] T. Fösel, P. Tighineanu, T. Weiss, and F. Marquardt, “Reinforcement learning with neural networks for quantum feedback,” *Physical Review X* 8, 031084 (2018). doi:10.1103/PhysRevX.8.031084.
- [19] V. V. Sivak et al., “Model-free quantum control with reinforcement learning,” *Physical Review X* 12, 011059 (2022). doi:10.1103/PhysRevX.12.011059.
- [20] S. Borah et al., “Measurement-based feedback quantum control with deep reinforcement learning for a double-well nonlinear potential,” *Physical Review Letters* 127, 190403 (2021). doi:10.1103/PhysRevLett.127.190403.
- [21] K. Lewin, *Principles of Topological Psychology* (McGraw–Hill, 1936). <https://books.google.com/books?id=IGB9AAAAMAAJ>.
- [22] J. B. Kruskal, “Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis,” *Psychometrika* 29, 1–27 (1964). doi:10.1007/BF02289565.
- [23] J. B. Tenenbaum, V. de Silva, and J. C. Langford, “A global geometric framework for nonlinear dimensionality reduction,” *Science* 290, 2319–2323 (2000). doi:10.1126/science.290.5500.2319.
- [24] N. E. Steenrod, *The Topology of Fibre Bundles* (Princeton University Press, 1951). doi:10.1515/9781400883875.
- [25] Y. Ollivier, “Ricci curvature of Markov chains on metric spaces,” *Journal of Functional Analysis* 256, 810–864 (2009). doi:10.1016/j.jfa.2008.11.001.
- [26] A. Einstein, “Die Grundlage der allgemeinen Relativitätstheorie,” *Annalen der Physik* 354, 769–822 (1916). doi:10.1002/andp.19163540702.